



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
12-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

Maximum Marks : 216

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

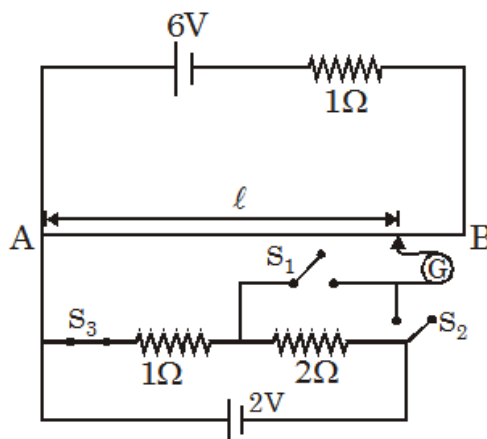
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

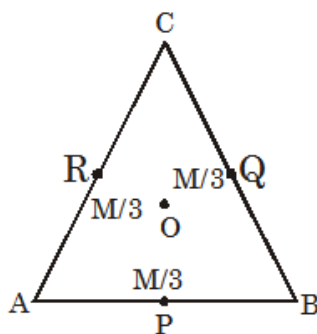
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Figure shows a potentiometer circuit. The length of potentiometer wire AB is 120 cm and its resistance is 2Ω . The internal resistance of 2V cell is 1Ω . Initially the switches S_1 and S_2 are open and S_3 is closed.



- (A) When only S_2 is open, balanced length $\ell = 20$ cm.
- (B) When only S_1 is open, balanced length $\ell = 60$ cm
- (C) When all three switches are closed, balanced length is 60 cm.
- (D) When S_3 is open, S_2 is also open but S_1 is closed, balance length is 60 cm.

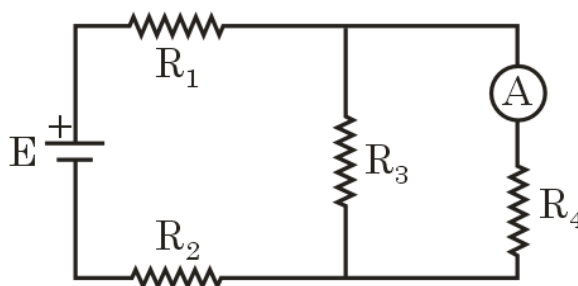
2. A thin rod of length ℓ is kept vertically (along y-axis) with end A being at rest on a smooth horizontal plane. A slight disturbance makes the end A slip on the plane (along x-axis) and the rod falls. Then,
- (A) the path described by a point P at a distance of $\frac{\ell}{4}$ from A is a quadrant of a circle.
- (B) the centre of mass of the rod moves along a straight line, parallel to y-axis.
- (C) the acceleration of the centre of mass as the rod becomes horizontal is $\frac{3g}{4}$.
- (D) the greatest angular velocity of the rod about the instantaneous centre of rotation is $\sqrt{\frac{3g}{\ell}}$.
3. A, B and C are the vertices of a uniform equilateral triangular thin lamina of mass M. Three particle of masses $\frac{M}{3}$, each are located at the midpoints P, Q and R of the edges of the lamina. For this system, (each edge of lamina = a)



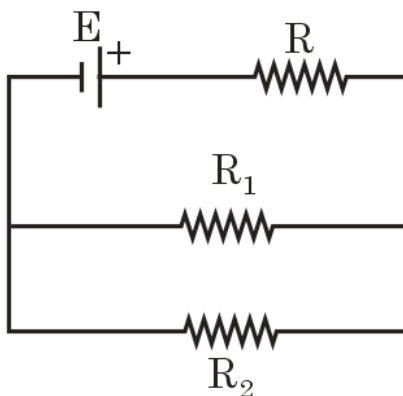
- (A) the moment of inertia of the system about an axis through the centroid O perpendicular to the plane is $\frac{Ma^2}{6}$.
- (B) the moment of inertia of the system about an axis through O perpendicular to the plane is $\frac{5Ma^2}{12}$
- (C) the moment of inertia of the system about an axis through A perpendicular to the plane is $\frac{13Ma^2}{12}$
- (D) the moment of inertia of the system about an axis through C perpendicular to the plane is $\frac{5Ma^2}{6}$

Space for Rough Work

4. A network of four unequal resistors R_1 , R_2 , R_3 and R_4 are connected in a network (see diagram) that contains a source of e.m.f. E and an ammeter A . The source has negligible internal resistance. For this network.



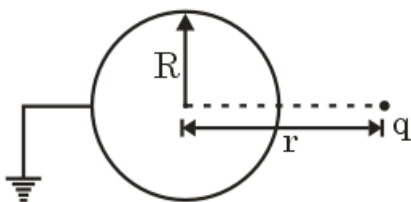
- (A) If $R_3 > R_4$, the current in the resistor R_3 will be less than that shown by the ammeter.
- (B) If $R_3 < R_4$, the current in the resistor R_2 will be greater than that shown by the ammeter.
- (C) If the source of e.m.f. and the ammeter were physically interchanged in the network the ammeter will show the same current.
- (D) If the resistors R_2 and R_1 were physically interchanged, the ammeter will show the same current
5. In the given network, the powers generated in the resistors R_1 and R_2 are P_1 and P_2 and the power generated in R is P . For this network,



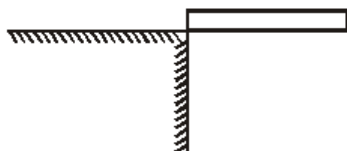
- (A) If $P_1 = P_2$, it necessarily follows that $R_1 = R_2$
- (B) If $R = R_1 + R_2$, then $P > P_1 + P_2$
- (C) If $P = P_1 + P_2$, then $R \leq \frac{R_1 + R_2}{4}$
- (D) If $R_1 = R_2$, then it necessarily follows that $P = P_1 + P_2$

Space for Rough Work

6. A charge q is placed at a distance r from centre of conducting sphere as shown in figure. Now sphere is earthed then correct options are :-



- (A) Induced charge on sphere is $-\left(\frac{R}{r}\right)q$
- (B) Induced charge on sphere is $-q$
- (C) Force acting between sphere and external charge q is $\frac{1}{4\pi\epsilon_0} \frac{q^2 R r}{(r^2 - R^2)^2}$
- (D) Force between sphere and external charge q is $\frac{q^2}{r^2}$.
7. A uniform disc of mass 2kg and radius 1m is mounted on a horizontal axle supported on fixed frictionless bearings. A light chord is wrapped around the rim of the disc and mass of 1kg is tied to the free end. If it is released from rest (take $g = 10\text{ m/s}^2$)
- (A) the tension in the chord is 5N .
- (B) in first four seconds the angular displacement of the disc is 40 rad .
- (C) the work done by the torque on the disc in first four seconds is 200 J .
- (D) the increase in the kinetic energy of the mass in the first four seconds is 200 J .
8. A uniform rod of length ℓ is falling down with a velocity v_0 when one of it's end hits a fixed edge as shown. (Consider impulsive force to be acting perpendicular to the length of rod)



- (A) If the collision is elastic, the centre of mass may have upwards velocity just after the collision.
- (B) If the collision is inelastic, the centre of mass may come to rest just after the collision.
- (C) If the collision is perfectly inelastic, the centre of mass has a downward velocity just after the collision.
- (D) If the collision is elastic, the centre of mass has a downward velocity just after the collision.

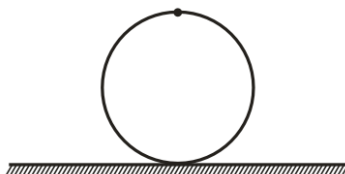
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

A small particle of mass m is fixed to the perimeter of a ring of same mass and radius ' r ' and ring is placed on a horizontal plane. Friction is negligible on horizontal plane. Initially particle is at top most point, then the system is released from rest. Answer next two questions when the particle is at the same height as the centre of ring after being released from topmost point. Assume that the ring stays in vertical plane during its motion under consideration.



9. Mark the correct option(s) :-

(A) Speed of centre of mass of system is $\sqrt{\frac{gr}{4}}$.

(B) Angular velocity of system is $\omega = \sqrt{\frac{g}{r}}$

(C) Contact point of ring with ground is instantaneously at rest.

(D) Centre of ring is instantaneously at rest.

10. Mark the correct option(s) :

(A) Instantaneous acceleration of the centre of ring is $\frac{g}{2}$.

(B) Acceleration of centre of ring initially points towards left and then towards right.

(C) The force exerted by the ring on the ground at the instant given is $\frac{3}{2}mg$.

(D) The force exerted by the ring on the ground at the instant given is $2mg$.

Space for Rough Work

Paragraph for Questions 11 and 12

A rocket is launched from and returns to a spherical planet of radius R in such a way that its velocity vector on return is parallel to its launch vector. The angular separation at the centre of the planet between the launch and arrival points is θ .

11. Which of the following can be the distance of the rocket above the surface of the planet ?
- (A) R (B) $\frac{R}{2}$ (C) $\frac{R}{3}$ (D) $4R$
12. How long does the flight of rocket take, if the period of satellite flying around the planet just above its surface is T_0 ?
- (A) $T_0 \left(\frac{1}{2} + \frac{1}{\pi} \right)$ if $\theta \rightarrow 0$ (B) $T_0 \left(\frac{1}{2} + \frac{1}{\sqrt{2}\pi} \right)$ if $\theta = 90^\circ$
- (C) $T_0 \left(\frac{1}{2} + \frac{2}{\pi} \right)$ if $\theta \rightarrow 0$ (D) $T_0 \left(\frac{1}{2} + \frac{\sqrt{2}}{\pi} \right)$ if $\theta = 90^\circ$

Space for Rough Work

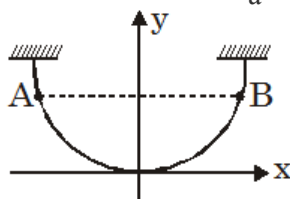
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

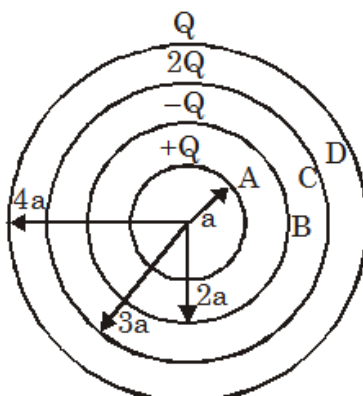
Zero Marks : 0 In all other cases.

1. A particle is uncharged and is thrown vertically upward from ground level with a speed of $5\sqrt{5}$ m/s. As a result, it attains a maximum height h . The particle is then given a positive charge $+q$ and reaches the same maximum height h when thrown vertically upward with a speed of 13 m/s. Finally, the particle is given a negative charge $-q$. Ignoring air resistance, determine the speed (in m/s) with which the negatively charged particle must be thrown vertically upward, so that it attains exactly the same maximum height h .
2. A smooth parabolic wire track defined by relation $y = \frac{x^2}{2a}$ is fixed in x - y plane in a gravity free space as shown in figure. An electric field $\vec{E} = \alpha y\hat{i} + \alpha x\hat{j}$ (where α is a constant) exist in the region. A small ring the mass ' m ', charge ' q ', which just fits in wire track, is released from rest at position A $\left(-a, \frac{a}{2}\right)$. Find speed of the ring when it reaches at point B $\left(a, \frac{a}{2}\right)$ if $q\alpha = \frac{2m}{a^2}$.

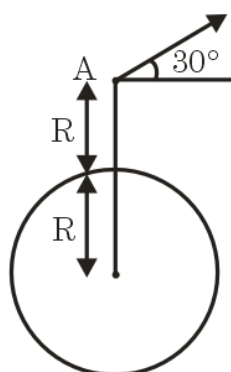


Space for Rough Work

3. Consider four concentric conducting spherical shells of radius a , $2a$, $3a$ and $4a$ having charges Q , $-Q$, $2Q$ and Q respectively as shown in figure. Total electrostatic potential energy of this charge configuration is $\left(\frac{nQ^2}{\pi\epsilon_0 a}\right)$. Find n .

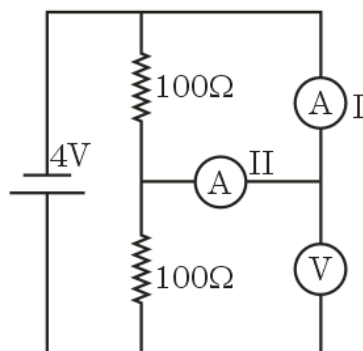


4. A voltmeter coil has resistance 50.0Ω and a resistance of $1.15 \text{ k}\Omega$ is connected in series. It can read potential differences up to 12 volts. If this same coil is used to construct an ammeter which can measure currents upto 2.0 A , what should be the resistance of the shunt used ?
5. A particle of mass m is fired from the point A (as shown in the figure) situated at a distance of $2R$ from the centre of the planet with a velocity $\frac{1}{\sqrt{2}}$ times the escape velocity from point A. The radius of the planet is R . The maximum distance of the particle from the centre in the subsequent motion is nR . Find the value of n .

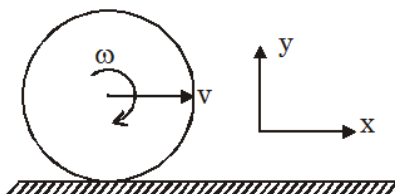


Space for Rough Work

6. A metal sphere of radius a is surrounded by a concentric metal spherical shell of radius b ($b > a$). The space between the spheres is filled with material whose electrical conductivity σ varies with the electric field strength E according to the relation $\sigma = KE$ where K is constant. A potential difference V is maintained between inner sphere and outer shell. If the current (in ampere) between the inner sphere and outer shell is x , then find $x/4$. [Here $K = \frac{1}{\pi} \times 10^{-2} \text{ ohm}^{-1} \text{ volt}^{-1}$, $V = 20$ volts, $b = ea$ ($e = \text{exponential}$) and $\pi = 22/7$]
7. In the figure ammeter (I) reads a current of 10 mA, while the voltmeter reads a potential difference of 3V. What does ammeter (II) (in mA) read ? The ammeters are identical, the internal resistance of the battery is negligible. (Consider all ammeters and voltmeters as non ideal)



8. A disc of radius '5cm' rolls on a horizontal surface with linear velocity $v = 1 \hat{i} \text{ m/s}$ and angular velocity 50 rad/sec. Height of particle from ground on rim of disc which has velocity in vertical direction is (in cm) :



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
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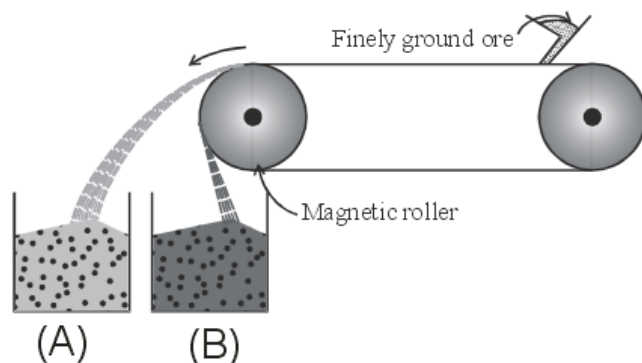
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1. For non-stoichiometric cuprous oxide, $\text{Cu}_{1.80}\text{O}_{1.00}$, select the correct information(s). (Cu = 63.5) :
 - (A) It behaves as p-type semiconductor
 - (B) % of total copper as Cu(I) is 11.11%
 - (C) It has metal deficiency defect
 - (D) Density of solid is 20% less than the ideal Cu_2O crystal
2. Liquids 'A' and 'B' form an ideal solution. At 25°C , $P_A^\circ = 25 \text{ cm Hg}$ and $P_B^\circ = 75 \text{ cm Hg}$. If 5.0 moles of liquid 'A' is mixed with 20.0 moles of liquid 'B' and the pressure over the system is maintained at 50 cm Hg, the correct information(s) regarding the final composition of system is/are :
 - (A) Vapour contains 5.0 moles of 'A'.
 - (B) Liquid solution contains 50% of 'A', by moles.
 - (C) Vapours contains 75% of 'B', by moles
 - (D) 50% of the total moles of liquid taken will vaporise.

Space for Rough Work

3.



If above setup is used during electro-magnetic separation of magnetic particles with non-magnetic particles then which of the following observation will be observed.

- (A) Most of the part of magnetic ore will be collected in container 'B'
 - (B) Silicates impurities are collected in container 'A'
 - (C) It is based on physical properties of matter
 - (D) In case of cassiterite, gangue particles are collected in container 'A'
4. Which of the following is/are correct for commercial extraction of metals?
- (A) During Matte formation FeO is removed in the form of slag
 - (B) Copper from it's low grade ore can be extracted by hydrometallurgical reduction using H_2 gas.
 - (C) Almost all Au is extracted by amalgamation process completely from Aluvial sand
 - (D) Metal sulphides are more stable as compared to corresponding metal oxides
5. Three complex compounds I, II and III are $[M(H_2O)_6]^{3+}$, $[M(en)_3]^{3+}$, $[M(CN)_6]^{3-}$ respectively. Complex I and III have green and red colour respectively then identify the correct possible matching colour for complex II.
- (A) Blue (B) Orange (C) Yellow (D) Violet

Space for Rough Work

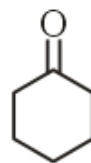
6. $(P) + NaHSO_3 \rightleftharpoons (Q)$

Compound (P) having molecular formula ' C_3H_6O ' & gives (+ve) haloform test. Choose the correct statement(s)?

(A) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad SO_3Na \end{array}$ and crystalline in nature

(B) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad SO_3Na \end{array}$ and non-crystalline in nature

(C) (P) can be aldehyde, methyl ketone & some cyclic ketones as



(D) (Q) is $\begin{array}{c} H_3C \quad \quad OH \\ \quad \diagdown \quad \diagup \\ \quad C \\ \quad \diagup \quad \diagdown \\ H_3C \quad \quad OSO_2Na \end{array}$ and crystalline in nature

7. $\text{Cyclohexene} \xrightarrow{OsO_4/H_2O} (A) \xrightarrow{HIO_4} (B) \xrightarrow[\Delta]{dil. OH^-} (C) \xrightarrow[(ii) Zn/H_2O]{(i) O_3} (D)$

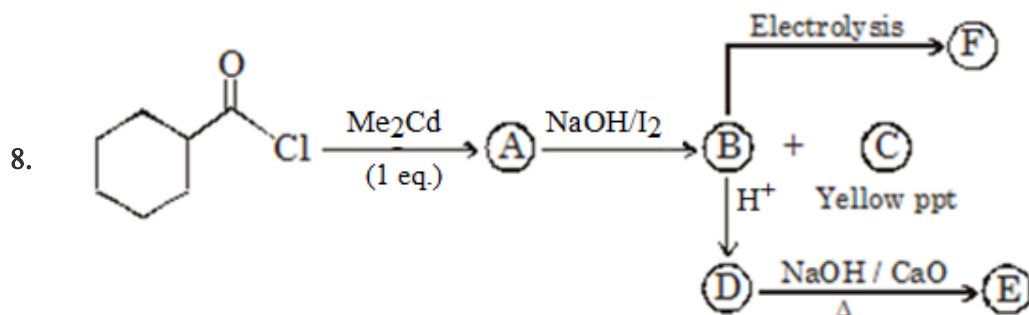
Correct statement(s) is/are

(A) (A) is vicinol diol

(B) (B) is a dicarboxylic acid

(C) (B) gives Tollen's test

(D) (D) gives Iodoform test



Correct statement is/are :

(A) Compound F is

(B) Compound E is

(C) A is acetophenone

(D) In reaction from A to B, 3 equivalent of I_2 consumed, with respect to 1 equivalent of A.

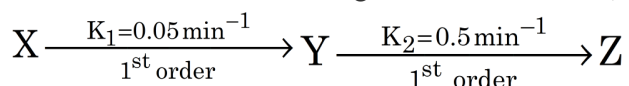
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

A substance 'X' converts into 'Z' through some intermediate, 'Y'.



t = 0	0.50M	0	0
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9. The correct concentration(s) of 'X', 'Y' and 'Z' at $t = 20$ min is ($e^{-1} = 0.36$, $e^{-10} = 4.5 \times 10^{-5}$) :
- (A) $[X] = 0.18$ M (B) $[Y] \approx 0.02$ M
- (C) $[Z] \approx 0.30$ M (D) $[X] + [Y] + [Z] = 0.50$ M
10. The correct information(s) is/are ($\ln 2 = 0.7$, $\ln 1.6$) :
- (A) The concentration of 'Y' becomes maximum at about 5.11 min. from the start of reaction
- (B) The rate of formation of 'Z' becomes maximum at about 5.11 min. from the start of reaction
- (C) The rate of disappearance of 'X' becomes equal to the rate of appearance of 'Z' at about 5.11 min. from the start of reaction
- (D) Concentration of 'X' becomes 0.25 M at about 5.11 min. from the start of reaction

Space for Rough Work

Paragraph for Questions 11 and 12

An octahedral homoleptic high spin complex of '3d' series central metal ion having $3d^n$ electronic configuration have $t_{2g}^x e_g^2$ arrangement of electrons.

Answer the following based on above paragraph.

11. Choose the correct pair of central atom/ion and ligand for the above mentioned complex.
 (A) Fe^{2+} , NH_3 (B) Co^{3+} , NH_3 (C) Fe^{3+} , F^- (D) Ni^{2+} , H_2O
12. For the maximum possible value of 'x' the value of spin only magnetic moment will be.
 (A) $\sqrt{8}$ B. M. (B) $\sqrt{15}$ B. M. (C) $\sqrt{24}$ B. M. (D) $\sqrt{35}$ B. M.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. N_2 and H_2 gases are mixed in 1 : 3 mole ratio. At equilibrium at 100 atm, the equilibrium mixture is found to contain 20 mole percent NH_3 . For the reaction :
$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
$$K_p \text{ is } \left(\frac{x}{216} \right) \text{ atm}^{-2}. \text{ The value of 'x' is.}$$
2. 2 moles of an ideal gas undergoes reversible isothermal expansion from 5 dm^3 to 20 dm^3 at 500 K. The amount of heat absorbed by the gas (in Kcal) is ($\ln 2 = 0.693$).
3. The standard enthalpies of combustion of C(graphite), $\text{H}_2(\text{g})$ and $\text{C}_3\text{H}_8(\text{g})$ are -393, -285 and -3280 kJ/mole, respectively. The standard enthalpy of formation (in KJ/mole) of $\text{C}_3\text{H}_8(\text{g})$ is.
4. If all the bond angles w.r.t. 'N' and 'B' are assumed as 120° in $\text{BH}_2(\text{NMe}_2)$ molecule then maximum number of atoms in one plane will be :-

Space for Rough Work

5. Some of the following metallurgical process are given below :-

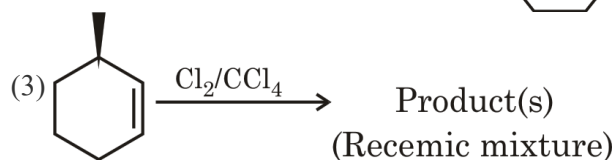
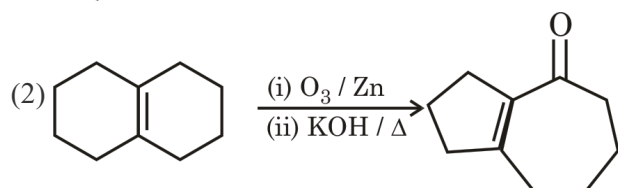
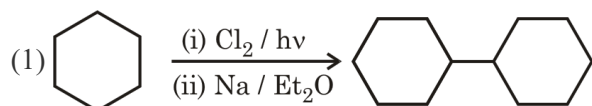
- a. Electro magnetic separation
- b. Froth floatation process
- c. Calcination
- d. Roasting
- e. Leaching
- f. Reduction by coke
- g. Self reduction
- h. Electrolytic refining
- i. refining of metal by liquation

The number of above stages involved in the extraction of Zn from Calamine and Cu from Chalcocite are X and Y. the value of $X + Y$ is.

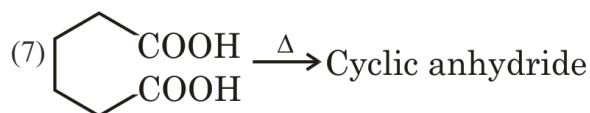
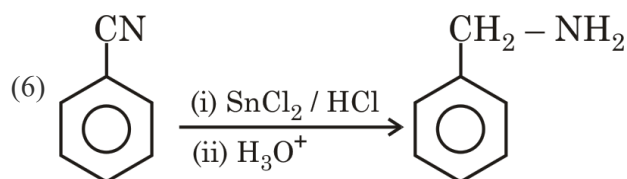
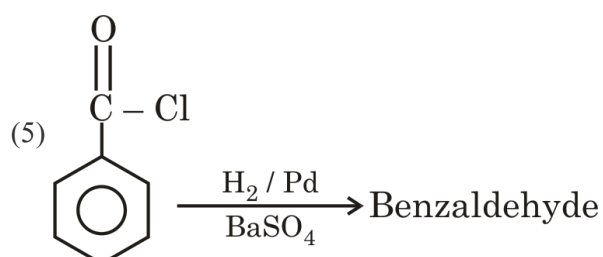
6. R-2-chlorohexane $\xrightarrow[h\nu]{Cl_2}$ Number of monochloro products.

Space for Rough Work

7. Find number of reaction correctly matched, with their major product (or statement written with them) :



(4) CH_3CHO gives positive Tollen's test as well as Lucas test



8. Which of the following compound when reacts with CH_3MgCl gives methane gas
(i) Tertiary butylchloride (ii) Tertiary butylalcohol (iii) Di-butylether (iv) Phenol
(v) Pyrole (vi) Pyridine

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The value of $\lim_{x \rightarrow \infty} x^2 \tan \left(\ln \left(\sqrt{\cos \frac{\pi}{x}} \right) \right)$ is equal to
 (A) $\frac{-\pi^2}{2}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{-\pi^2}{4}$ (D) $\frac{\pi^2}{4}$
2. If $\int \frac{x^3 - 8}{x^3 \sqrt{x^4 - x^3 + 2}} dx = \frac{k \cdot \sqrt{x^4 - x^3 + 2}}{x^2} + C$, then the value of k is - (where 'C' is integration constant)
 (A) 3 (B) 2 (C) 7 (D) 4
3. If $\int \frac{3z^3 - 8z + 5}{\sqrt{z^2 - 4z - 7}} dz = (z^2 + az + 36) \sqrt{z^2 - 4z - 7} + b \ln |z - 2 + \sqrt{z^2 - 4z - 7}| + C$, where $a, b \in \mathbb{I}$ and C is integration constant, then -
 (A) $a > b$ (B) $a < b$
 (C) $a + b = 117$ (D) exactly one out of a or b is a prime number

Space for Rough Work

4. Let $f(x) = \cos^{-1} \left(\ln \left(\frac{x+1}{x} \right) \right)$, then which of the following statement(s) is/are correct
- (A) Domain of $f(x)$ contains $(-\infty, -1) \cup (1, \infty)$
- (B) Range of $f(x)$ contains $\left[0, \frac{1}{e-1} \right]$
- (C) Domain of $f(x)$ is $\left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$
- (D) Range of $f(x)$ is $(-\infty, -1] \cup (1, \infty)$
5. Let $f(x)$ be a non constant twice derivable function defined on $\mathbb{R}$ such that $f(2+x) = f(2-x)$ and $f' \left(\frac{1}{2} \right) = 0 = f'(1)$, then minimum number of roots of the equation $f''(x) = 0$ in $(0, 4)$, is
- (A) 4 (B) 3 (C) 2 (D) 5

Space for Rough Work

6. Let $F(n, x) = \sum_{r=1}^n \tan^{-1} \left(\frac{2x}{4 + r(r+1)x^2} \right)$, where $x > 0$ and $n \in \mathbb{N}$ and $\frac{d}{dx} F(n, x) = G(n, x)$, then
- (A) $F(8, 2) = \tan^{-1} \left(\frac{4}{5} \right)$ (B) $\lim_{n \rightarrow \infty} F(n, 2\sqrt{3}) = \frac{\pi}{6}$
- (C) If $G\left(n, \frac{1}{3}\right) < 0$, then $n \geq 36$ (D) $\lim_{n \rightarrow \infty} G(n, 1) = -\frac{2}{5}$
7. If $a + b + c = 4$ and $a^2 + b^2 + c^2 + 3(ab + bc + ac) = 21$ where $a, b, c \in \mathbb{R}$, then which of the following is/are true
- (A) minimum value of $(a+b)(b+c)(a+c)$ is 18
- (B) minimum value of $(a+b)(b+c)(a+c)$ is 24
- (C) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{490}{27}$
- (D) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{471}{27}$
8. Let $f(x) = \begin{cases} x[x] & -4 \leq x < -1 \\ 1-x & -1 \leq x < 1 \\ x^2 - 3x + 2 & 1 \leq x < 2 \\ (x-2)[x] & 2 \leq x \leq 4 \end{cases}$, then (where $[.]$ denotes greatest integer function)
- (A) number of points where $f(x)$ is non-differentiable is 6
- (B) sum of global maximum value and global minimum value of $f(x)$ is $\frac{63}{4}$
- (C) number of points of extrema of $f(x)$ is 3 points
- (D) number of solutions of $f(x) = 4$ is 3

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

Let $f(x) = \frac{1}{2} \left[f(xy) + f\left(\frac{x}{y}\right) \right]$ for $x, y \in \mathbb{R}^+$ such that $f(1) = 0$, $f'(1) = 2$ then

9. $f(x) - f(y)$ is
- (A) $f\left(\frac{y}{x}\right)$ (B) $f\left(\frac{x}{y}\right)$ (C) $f(2x)$ (D) $f(2y)$
10. $f'(3)$ is greater than or equal to
- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Space for Rough Work

Paragraph for Questions 11 and 12

Consider a function $f(x) = ax^3 + x^2 - (2a + 1)x + 2$ (where $a \geq 0$)

11. If the number of value(s) of 'a' for which Rolle's theorem can be applied in the interval $[0, 1]$ is equal to k, then k is less than or equal to
- (A) 0 (B) 1 (C) 2 (D) 4
12. If k_1 and k_2 ($k_1 \neq k_2$) are respectively the local maximum and local minimum values of $f(x)$, then the sum of those roots of the equations $f(x) = k_1$ and $f(x) = k_2$ at which $f'(x) \neq 0$, is
- (A) $\frac{-2}{3a}$ (B) $\frac{-1}{a}$ (C) $\frac{-2}{a}$ (D) $\frac{-1}{3a}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int \frac{x^{17}}{(1+x^2)^{10}} dx = \frac{1}{n} \left(\frac{x^2}{1+x^2} \right)^m + C$ where $m, n \in \mathbb{N}$ and C is constant of integration, then value of $\frac{m}{n}$ (in its lowest form) is equal to
2. A circle of radius $2\sqrt{10}$ cm is divided by chord of length 12cm into two parts. If maximum area of rectangle inscribed into smaller circular segment is λ , then $\frac{\lambda}{\sqrt{15}}$ is
3. If the range of the function $y = |f(x)|$, where $f(x) = \frac{195x(x^2+4)}{9x^4+97x^2+144}$ is $\left[0, \frac{a}{b}\right]$ where a and b are relatively prime natural numbers then the value of $a - 4b$ is
4. If $\int \frac{dx}{(\sec x + \operatorname{cosec} x + \tan x + \cot x)^2} = \frac{x}{\alpha} + \frac{\cos(x + \frac{\pi}{4})}{\beta} + \frac{\cos 2x}{\gamma} + \delta$, then $|\alpha + \beta + \gamma|$ is
5. If $f(x) = \max\{\min(x, x^2), \max(x, x^2)\}$, where $x \geq 0$ then $\frac{d}{dx} \left((f \circ f)^{-1}(x) \right)$ at $x = 16$ is
6. The value of $\lim_{\theta \rightarrow 0^+} (\sin \theta)^{(\sin \theta - \sin^2 \theta)}$ is equal to
7. If $\frac{d^3 y}{dx^3} = - \left(\frac{dy}{dx} \right)^2 \left[k \left(\frac{d^2 x}{dy^2} \right) \cdot \left(\frac{d^2 y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 \left(\frac{d^3 x}{dy^3} \right) \right]$, then k is equal to
8. If $d = (x_2 - x_1)^2 + (-x_2 - 4 + 2\sqrt{x_1 + 2})^2$ where $x_1 > -2, x_2 \in \mathbb{R}$, then the minimum value of d is equal to -

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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